

Endogenous Elasticities: Price Multipliers Are Smaller for Larger Demand Shocks

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Increased interest in price impact at low frequencies

- ▶ Increased interest in **quantity questions** in asset pricing:
 - ▶ Holdings constant fundamental value of a security/portfolio, what is the **quantitative** impact of buying 1% of outstanding supply?
 - ▶ Previously a microstructure question (execution cost of \$1 trade).
Also (perhaps more) interesting at longer frequencies and \$1B trades.
- ▶ The typical **linear** regression: $\Delta p = M * \Delta d + \epsilon$.
 - ▶ Δp : Returns in (%)
 - ▶ Δd : Shares bought/sold as fraction of supply (%).
 - ▶ Literature² finds $M \sim \mathcal{O}(1)$
- ▶ **Linearity is just an assumption.** Different models imply different relationships between Δp and Δd .³

² Index Inclusions/Reconstitutions (Pavlova & Sikorskaya, 2022) Dividend Reinvestments (Salomon & Hartzmark, 2022) Mutual fund fire sales (Coval & Stafford, 2008) Flow-driven trades (Lou 2012) GIV (Gabaix & Koijen, 2023) Ratings-driven Demand (Ben-David et al, 2024)

³E.g. Chacko, Jurek & Stafford (JF, 2007) microfound $\sqrt{\Delta d}$.

This paper

1. Test relationship between shock size $|\Delta d|$ and M

$$\Delta p = M_1 * \Delta d + M_2 * (|\Delta d| * \Delta d) + \epsilon$$

for 3 different demand shocks Δd

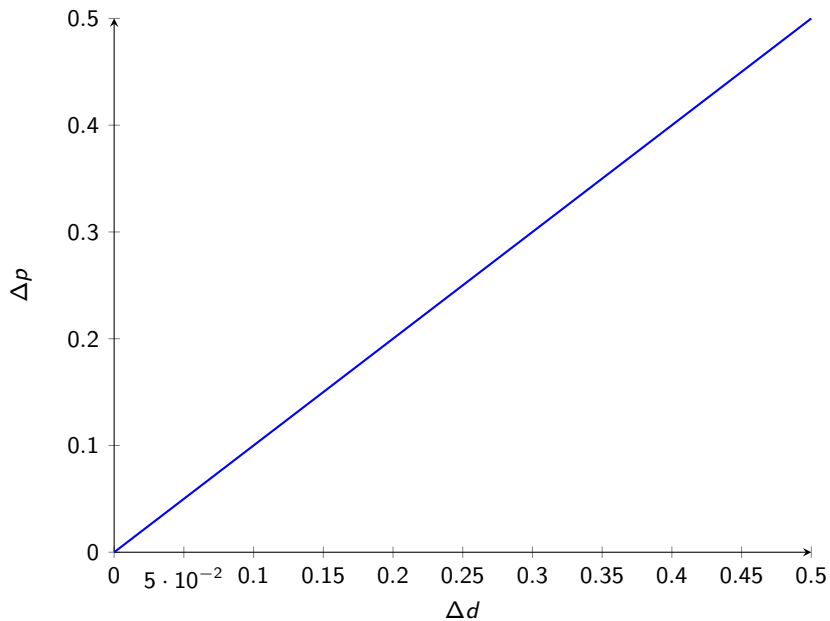
- ▶ Mutual fund flow-induced trades
- ▶ Index Reconstitutions
- ▶ Order-Flow Imbalance

2. Finding: M_2 is significantly negative. **Concave Price Impact**

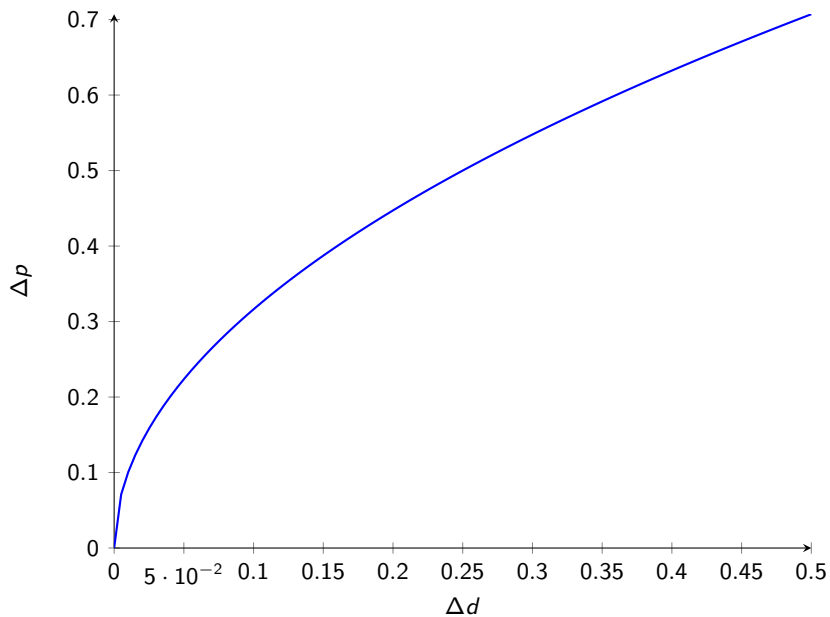
3. Interpretation:

- ▶ **Not** driven by: Heterogenous liquidity, time variation in M , endogenous liquidity taking
- ▶ Proposition: Endogenous slow-moving capital, declining marginal transaction costs, or costly information acquisition

Linear Price Impact



Concave Price Impact



Comments

► **Very nice paper!**

1. Well-known in industry that **transaction cost** are concave at higher frequency. Does this hold for low frequency shocks? Yes!
2. **Very** robust empirical finding. Allows distinguishing between different theories on pricing of demand shocks.
3. Important contribution to a fast-growing literature.

► **Comments**

1. Provide closer link to transaction costs.
2. What is the source of the concavity?
3. Cleaning up the cross-section of multiplier estimates.

T-costs: What's the price impact of buying ΔShares ?

- ▶ Price impact (in %) is larger when trading a larger fraction of supply
- ▶ Different approaches in the literature

1. Low frequency “Multiplier”:

- ▶ Assumes linearity, Supply = Shares Outstanding.

$$\Delta p = M \times \left(\frac{\Delta \text{Shares}}{\text{Shrout}} \right)$$

2. High frequency “Transaction Costs”

- ▶ Estimates power law $\alpha = \frac{1}{2}$, Supply = Avg. Daily Volume,

$$\Delta p = M \times \sigma \sqrt{\frac{\Delta \text{Shares}}{\text{Volume}}}$$

- ▶ E.g. ITG (now Virtu) cost model

Barra (1997), Kahn (1993), Grinold and Kahn (1999), and Loeb (1983), Chacko et al. (2007), Gabaix et al. (2006, 2007), Kyle & Obizhaeva (2016), Davila et al (2024)....

- ▶ Both literatures find that M is $\mathcal{O}(1)$
- ▶ Fun Fact: $\text{Shrout} \approx \frac{\text{Avg. Daily Volume}}{\sigma}$

Relationship to transaction costs

- ▶ Other than frequency (nanosecond/daily vs. monthly/quarterly) the two disjoint literatures on **multipliers** and **transaction costs** have a lot in common.
- ▶ Without explicit stance on origins of impact there is nothing special about the frequency.
 1. A large demand shift observed at a quarterly frequency consists of smaller trades executed at a higher frequency.
 2. Impacts should (at least the permanent component) add up! Do they?

Relationship to transaction costs

- ▶ Example using real data: APPL
 - ▶ Shrout 15B, Avg Daily Volume 50M ($\approx 100\%$ annual turnover).
- ▶ Assume that some uninformed trader buy 1% of shrout over 1 quarter (63 Trading days).
 - ▶ Implies buying $\frac{15B * 1\%}{63\text{Days}} = 2.4M$ Shares every day.

1. Daily T-cost (permanent part):⁴ $10bp * \sqrt{\frac{2.4M}{50M}} = 2.2bp/\text{Day}$
Total T-costs over quarter: $2.2bp/\text{Day} * 63\text{Days} = \mathbf{1.4\%}$

2. Multiplier-implied impact from this paper:
 $1\% * (2.3 - |1\%| * 10) = \mathbf{2.2\%}$

- ▶ Permanent component of daily T-costs adds up to long run multipliers (roughly).
- ▶ Amazing!

⁴10bp permanent price impact estimate from Frazzini et al. (2018)

What is the Source of the Concavity

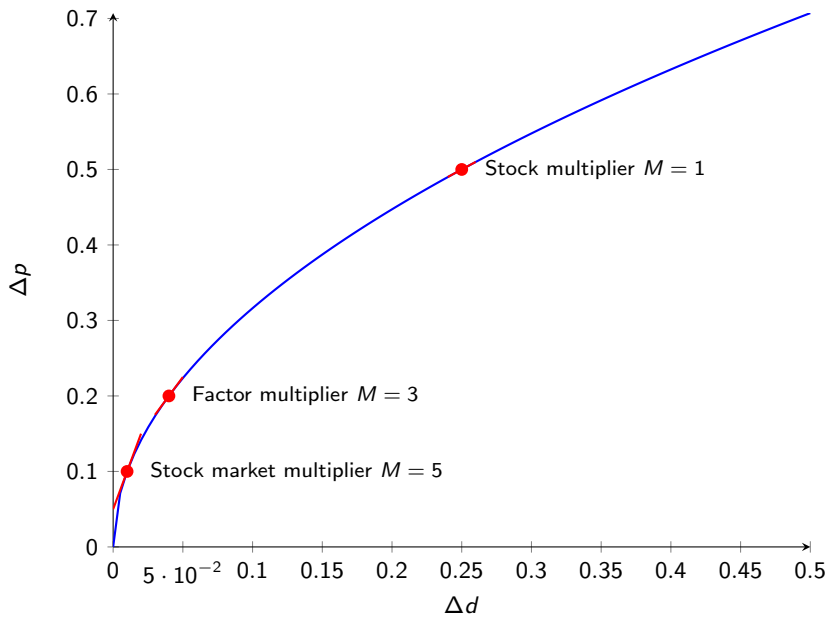
- ▶ A cool micro-foundation for concavity based on the Merton Model: Chacko, Jurek, Stafford (JF, 2007): “The Price of Immediacy”
- ▶ You show what it is not: Heterogenous liquidity, time variation in multipliers, endogenous liquidity taking. **But then what is it?**
- ▶ You propose
 1. Endogenous slow-moving capital (Duffie & Strulovici, 2012)
 2. Declining marginal transaction costs (Lo et al., 2004)
 3. Costly information acquisition (Nieuwerburgh & Veldkamp, 2009)
- ▶ Some of these can be tested explicitly!

What is the Source of the Concavity

- ▶ For every demand shock Δd at the quarterly level, you observe who provides liquidity from **13F position changes**.
- 1. Slow-Moving Capita/Investor Entry:
 - ▶ Do you see more extensive margin buys (investor entry) following large sales relative to small sales?
- 2. Decreasing Marginal Costs:
 - ▶ Your demand system gives you investor-specific rebalancing costs.
 - ▶ Stocks predominantly held by investors with decreasing rebalancing costs should have stronger concavity in their impact.
 - ▶ Could give additional justification for demand system approach.
- 3. Costly information acquisition:
 - ▶ Difficult to test.
 - ▶ Perhaps: Easier to “buy” incremental information for less covered (small cap) stocks? Concavity more pronounced for those stocks?

Price Multipliers are Larger at More Aggregated Levels

- ▶ “Price Multipliers are Larger at More Aggregated Levels” (Li & Lin, 2025)
- ▶ Can heterogenous demand shock sizes explain part of this effect?
 - ▶ For individual stocks: demand shocks (in % of supply) are larger
 - ▶ For factors: demand shocks (in % of supply) are smaller
 - ▶ For whole market: demand shocks (in % of supply) are tiny?



Conclusion

- ▶ Very nice paper!
 - ▶ Contributes to our understanding of **multipliers**, which are an important new moment for testing asset pricing models.
 - ▶ Intuitive and **very robust** finding: Multipliers are smaller for larger demand shocks.
- ▶ Small suggestions
 - ▶ Explicit link to T-costs.
 - ▶ What's the right exponent α that practitioners/academics should use at low frequency $M * (\Delta d)^\alpha$. Also square root $\alpha = 1/2$?
 - ▶ Test some of the proposed mechanisms.